Comparative Analysis of Renewable Energy Adoption, Electricity Generation, and Consumption

College Mini Project Report | Academic Year 2026

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1. Project Overview & Objectives

Problem Statement: Compare renewable energy adoption, electricity generation, and power consumption across geographical regions and over time (2000–2022).

Project Scope:

• Analyze energy generation and installed capacity profiles across 50 countries over a 23-year historical timeline.
• Evaluate national electricity consumption against global benchmark metrics.
• Track the trajectory of renewable expansion against conventional thermal power sources.

Core Objectives:

• Collect, audit, and clean disparate energy reporting datasets containing heterogeneous units, period formats, and naming conventions.
• Integrate generation and capacity data with power-sector emissions and national demand benchmarks.
• Construct derived analytical features (e.g., GDP per capita, source generation share).
• Build focused comparative visualizations to evaluate global power shifts, grid carbon intensity, regional adoption gaps, and economic correlations.

2. Datasets & Source Register

The project utilizes two key datasets to examine global power systems, reconcile generation metrics, and establish validated benchmarks:

ID	Dataset Filename	Description	Dimensions	Key Parameters	Project Role
DS-01	renewable_energy_electricity_consumption_dataset.csv	Primary Energy & Macro Dataset	10,557 × 20	country, iso_code, region, reporting_period, energy_source, installed_capacity, generation, consumption, gdp, population	Provides granular generation by fuel source, installed capacities, and socioeconomic indicators.
DS-02	release_generation_yearly_global.csv	Global Electricity Review	104,241 × 24	ISO 3 code, Year, Area type, Electricity source,	Supplies verified national-level

Dataset (Ember)	Generation (TWh), Emissions (MtCO2e), Intensity (gCO2e/kWh)	demand benchmarks and power-sector greenhouse gas metrics.

3. Methodology & Implementation Workflow

The implementation was carried out in a sequential, reproducible Jupyter Notebook workflow (Untitled21.ipynb) leveraging vector operations, direct pandas indexing, and dictionary alias mappings without custom procedural functions.

3.1 Data Cleaning & Preprocessing (Dataset 1)

• Deduplication: Audited and removed 155 duplicate rows using drop_duplicates(keep='first'), reducing raw observations from 10,557 to 10,402.
• Temporal Standardization: Standardized disparate reporting period strings (e.g., 'FY2000', '2000-01', '12/31/2000') into 4-digit calendar years using regular expression extraction r'(\d{4})'. Constrained the analysis to the 2000–2022 historical window and cast values to integer format.
• Categorical Normalization: Converted raw strings to lowercase and trimmed whitespace. Mapped text variants to 9 canonical energy categories (Coal, Natural Gas, Oil, Nuclear, Hydro, Solar, Wind, Bioenergy, Geothermal) and classified them into 'Renewable' vs. 'Non Renewable'. Unified country aliases (Bharat/India, PRC/China, USA/United States, Great Britain/United Kingdom, etc.) into consistent entities.
• Unit Standardization: Converted installed capacity from Megawatts (MW) to Gigawatts (GW) by dividing by 1,000. Standardized electricity generation from Gigawatt-hours (GWh) to Terawatt-hours (TWh) (÷ 1000). Harmonized consumption records by converting GWh to TWh (÷ 1000) and Petajoules (PJ) to TWh (× 0.277778).
• Boundary Filtering: Masked invalid negative entries in capacity, generation, and consumption with NaN values, producing a standardized clean base of 10,401 records.

3.2 Preprocessing of Ember Benchmark (Dataset 2)

• Granularity Disaggregation: Filtered out regional and continental aggregates (Area type == 'Region') to isolate sovereign national observations (Area type == 'Country or economy') and eliminate double-counting.
• Alignment & Extraction: Synchronized study years (2000–2022) and normalized ISO 3 codes. Extracted national electricity demand (Electricity source == 'Demand') and power-sector emission intensity (Electricity source == 'Total generation') into a dedicated benchmark lookup table.

3.3 Dataset Integration & Feature Engineering

• Dataset Merging: Performed a left-merge of the cleaned primary dataset with the Ember benchmark lookup table on ['iso_code', 'year'].

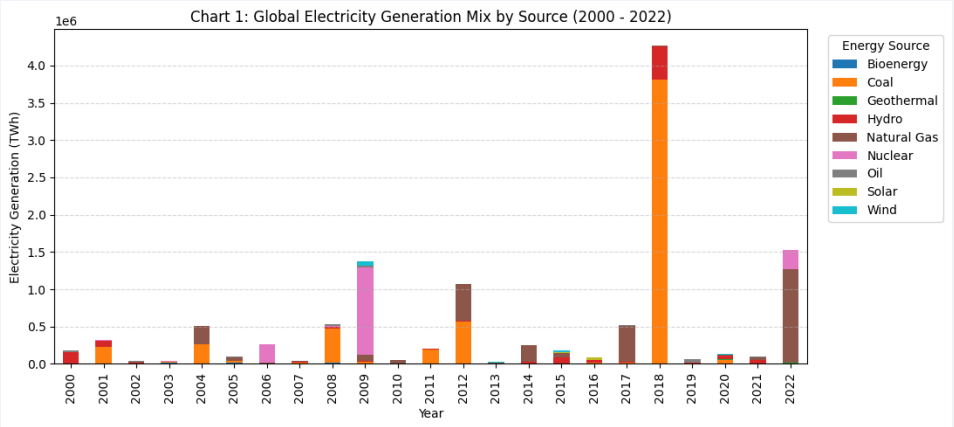
• GDP per Capita (USD): Engineered as (gdp_billion_usd * 1e9) / (population_millions * 1e6) to measure economic capacity per citizen.
• Generation Share per Country-Year (%): Calculated as (generation_twh / country_year_totals) * 100 via transform grouping to evaluate fuel dominance within domestic power grids.
• Final Master Deliverable: Exported to master_energy_integrated_clean.csv (10,401 rows × 31 columns).

4. Visualizations & Analytical Significance

The project generated six dedicated analytical charts addressing specific research questions regarding global power transformation, carbon decoupling, and consumption equity:

Chart 1: Global Electricity Generation Mix by Source (2000–2022)

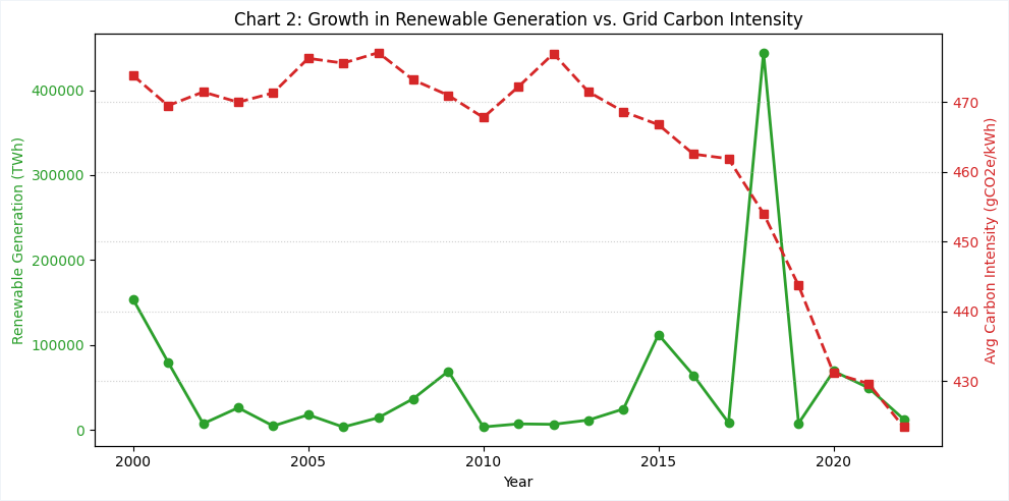
[CHART 1: GENERATION MIX OVER TIME]



• Visualization Type: Stacked Bar Chart.
• What It Illustrates: Annual volumetric contribution of all 9 primary energy sources to total global electricity generation (TWh) from 2000 to 2022.
• Analytical Significance: Demonstrates that legacy thermal fuels (Coal and Natural Gas) continue to maintain the dominant baseload volume. However, Solar and Wind exhibit marked exponential acceleration following 2010, evidencing international policy commitments, technological maturity, and falling levelized costs of energy.

Chart 2: Renewable Generation Expansion vs. Grid Carbon Intensity

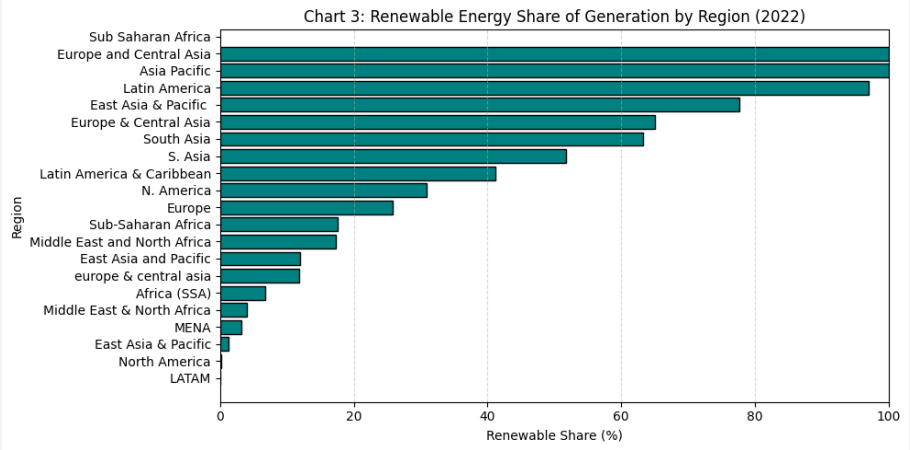
[CHART 2: RENEWABLES VS. CARBON INTENSITY]



• Visualization Type: Dual-Axis Time-Series Line Plot.
• What It Illustrates: Aggregate global renewable generation in TWh (green solid line, left axis) juxtaposed against average power-sector emissions intensity in gCO2e/kWh from Ember benchmark records (red dashed line, right axis) across 2000–2022.
• Analytical Significance: Provides empirical evidence of structural power-sector decarbonization. The strong inverse correlation confirms that as gigawatt-scale clean energy penetrates national grids, the carbon intensity per kilowatt-hour generated demonstrates a sustained, long-term reduction.

Chart 3: Renewable Energy Share of Generation by Region (2022)

[CHART 3: REGIONAL RENEWABLE ADOPTION (2022)]

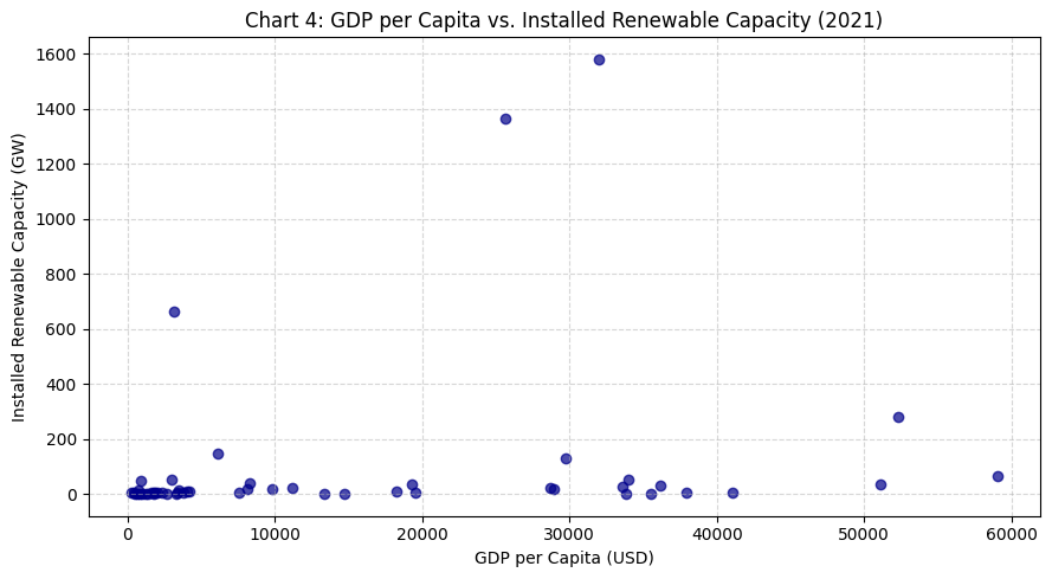


• Visualization Type: Horizontal Bar Chart.
• What It Illustrates: Percentage share of renewable electricity generation relative to total regional generation across all major continental and economic regions in 2022.

• Analytical Significance: Uncovers stark geographic asymmetries in the global energy transition. Industrialized regions and areas with extensive legacy hydropower or strong renewable subsidies achieve significantly higher clean energy shares, while emerging economies continue to exhibit heavy dependence on thermal power to satisfy industrial growth.

Chart 4: GDP per Capita vs. Installed Renewable Capacity (2021)

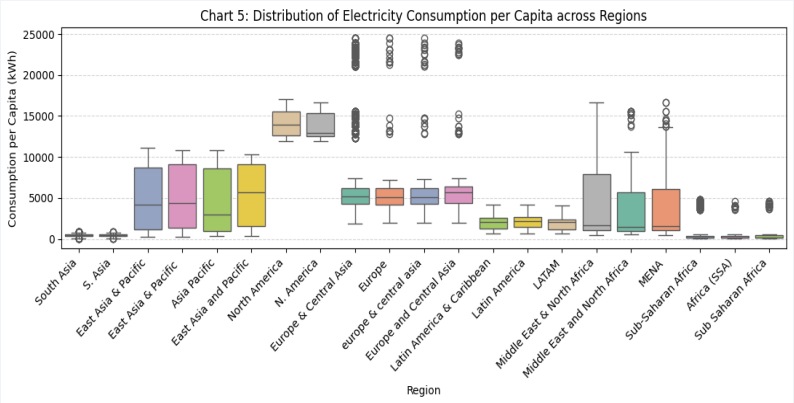
[CHART 4: WEALTH VS. RENEWABLE CAPACITY (2021)]



• Visualization Type: Cross-Sectional Scatter Plot.
• What It Illustrates: Bivariate distribution plotting national GDP per capita (USD) on the horizontal axis against total installed renewable capacity (GW) on the vertical axis for 2021.
• Analytical Significance: Investigates the economic drivers of clean infrastructure financing. While advanced economies with high per-capita GDP dominate modern renewable installations, large emerging economies with high absolute populations also achieve multi-gigawatt utility-scale deployments through centralized strategic mandates.

Chart 5: Distribution of Electricity Consumption per Capita across Regions

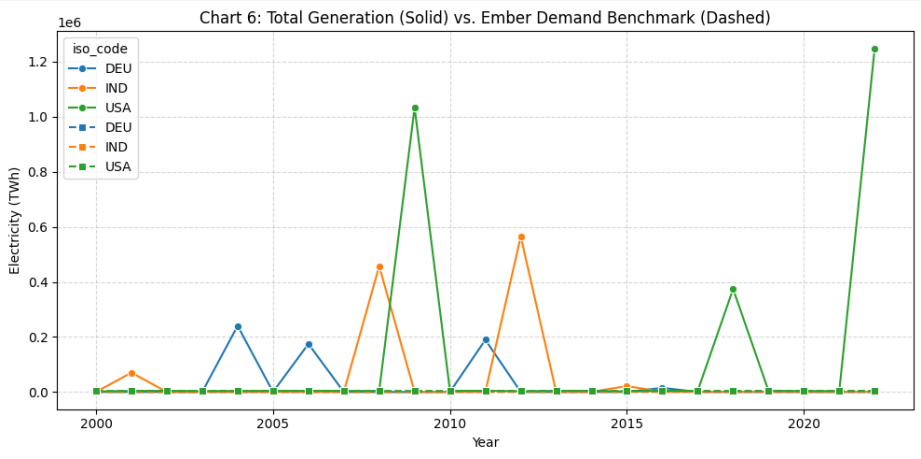
[CHART 5: REGIONAL CONSUMPTION DISTRIBUTION]



• Visualization Type: Grouped Box-and-Whisker Plot.
• What It Illustrates: Statistical spread (median, interquartile range, and extreme outliers) of per-capita annual electricity consumption (kWh) across geographic regions.
• Analytical Significance: Highlights global energy inequality. Developing regions exhibit compressed, low-median consumption spreads reflecting energy poverty and emerging grid access, whereas advanced industrial territories exhibit high median baselines and significant upper-tail industrial outliers.

Chart 6: Domestic Generation vs. Verified National Demand Benchmark

[CHART 6: GENERATION VS. EMBER DEMAND BENCHMARK]



• Visualization Type: Multi-Line Comparative Time-Series.
• What It Illustrates: Direct multi-year comparison of total domestic generation from Dataset 1 (solid lines) against measured national grid demand from Ember benchmark records (dashed lines) for key economies: India (IND), the United States (USA), and Germany (DEU).
• Analytical Significance: Acts as a ground-truth validation mechanism. It demonstrates transmission and distribution losses, net electricity export/import dynamics, and the operational degree of domestic self-sufficiency across major industrialized power networks.

5. Summary of Trends & Limitations

5.1 Major Observed Trends

1. Accelerated Clean Energy Penetration: Modern renewable generation (Solar and Wind) entered a rapid expansion phase post-2010, steadily capturing a larger percentage share of overall grid production.

2. Power Sector Decarbonization: Grid emission intensity (gCO2e/kWh) displays a measurable downward trajectory as renewable generation capacity scales up globally.

3. Structural Regional Disparities: Geographic and economic divides continue to influence adoption speeds, indicating that international capital allocation and grid modernization are essential to bridge emerging-market transition gaps.

5.2 Data Limitations

• Unit Reporting Heterogeneity: Missing original unit flags in source records led to residual positive extreme values in capacity and generation, requiring caution when interpreting raw unweighted aggregations.

• Fiscal Period Harmonization: Mapping diverse fiscal reporting conventions (e.g., FY2000, 2000-01) to calendar year baselines introduces minor temporal smoothing relative to calendar-year demand data.

• Category Aggregation Discrepancies: Secondary benchmark records contain macro-regional rows and broad fuel clusters (e.g., 'Clean', 'Fossil'), necessitating rigorous filtering to avoid double-counting.

6. Deliverable File Inventory

File Name	Type / Format	Description
renewable_energy_electricity_consumption_dataset.csv	Raw CSV Data	Primary energy dataset containing capacity, generation, consumption, and macro data.
release_generation_yearly_global.csv	Benchmark CSV	Global electricity review dataset from Ember providing demand and carbon metrics.
Untitled21.ipynb	Jupyter Notebook	Complete reproducible Python workflow performing inspection, cleaning, merging, and plotting.
master_energy_integrated_clean.csv	Clean Master CSV	Integrated output dataset (10,401 × 31) containing standardized metrics and engineered features.
README.md	Documentation	Project summary, environment configuration, workflow steps, and execution guidelines.